

COMSATS University Islamabad

Attock Campus



Department Of Computer Science

<i>Course</i>	<i>Information Security (Lab)</i>
<i>Instructor</i>	<i>Ms. Ambreen Gul</i>
<i>Assignment No.</i>	<i>01</i>

Student Details

<i>Registration No.</i>	<i>Name</i>
<i>FA24-BSE-039</i>	<i>Muhammad Taha Niaz</i>

Ceaser Cipher

Code:

```
1  def caesar_encrypt(text, shift): 2 usages
2      result = ""
3
4      for char in text:
5          # Check if character is uppercase letter
6          if char.isupper():
7              new_char = chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
8              result += new_char
9
10         # Check if character is lowercase letter
11         elif char.islower():
12             new_char = chr((ord(char) - ord('a') + shift) % 26 + ord('a'))
13             result += new_char
14
15         # If not a letter (space, number, symbol), keep it same
16         else:
17             result += char
18
19     return result
20
21 def caesar_decrypt(ciphertext, shift): 1 usage
22     return caesar_encrypt(ciphertext, -shift)
23
24 # ---- Main Program ----
25 message = input("Enter your message: ")
26 shift_value = int(input("Enter shift value: "))
27
28 encrypted = caesar_encrypt(message, shift_value)
29 print("Encrypted Message:", encrypted)
30
31 decrypted = caesar_decrypt(encrypted, shift_value)
32 print("Decrypted Message:", decrypted)
```

Explanation:

This program implements the Caesar Cipher, a method of encrypting text by shifting each letter by a fixed number of positions in the alphabet.

Encryption Function

The function `caesar_encrypt(text, shift)`:

- Takes a message and a shift value as input.

- Loops through each character in the text.
- If the character is a letter:
 - It converts it to its ASCII value using `ord()`.
 - Shifts it using the formula with `% 26` to stay within alphabet range.
 - Converts it back to a character using `chr()`.
- If the character is a space or special symbol, it remains unchanged.
- Returns the encrypted message.
- Maintains uppercase and lowercase letters.

Decryption Function

The function `caesar_decrypt(ciphertext, shift)`:

- Calls the encryption function with a negative shift value.
- This reverses the encryption and restores the original message.